Computer Networks
EXP 7:Wireshark Statistics and Graphs

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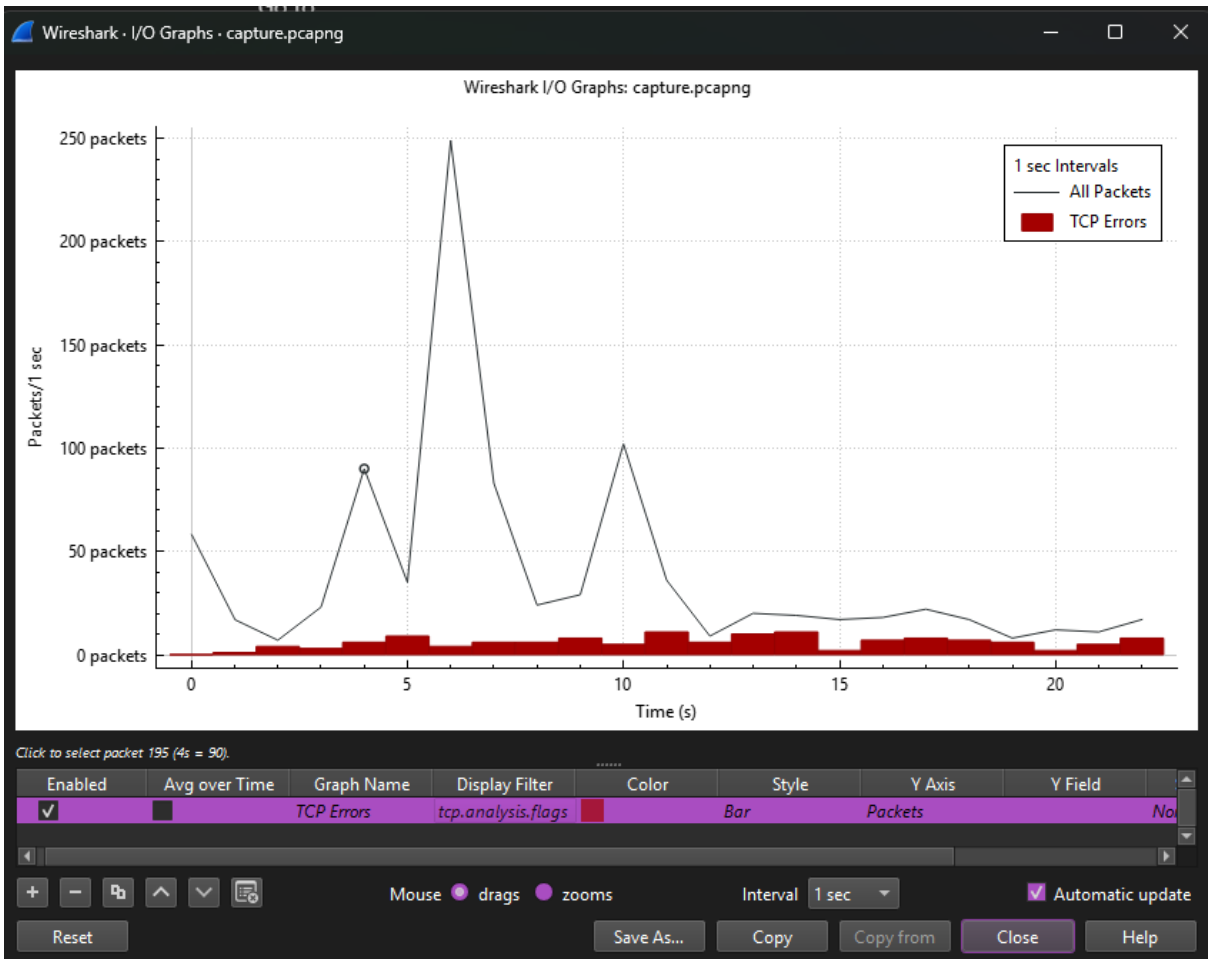
IO GRAPHS:

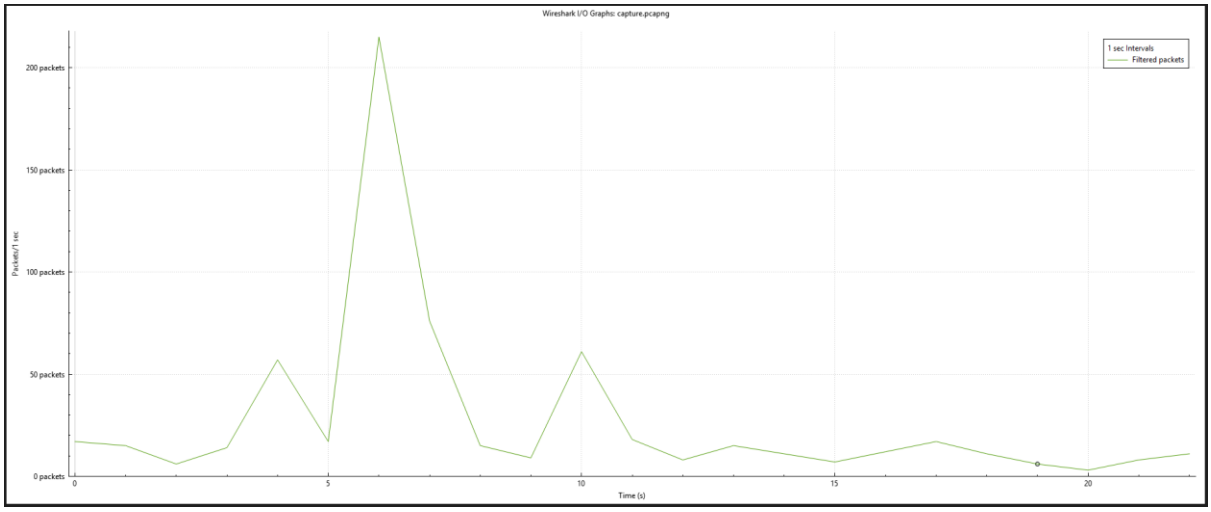
1.Total packets over time:



Inferences:

1. Packet rate increases when browsing activity starts.
2. Peak traffic observed during video streaming.
3. Idle periods show very low packet rate.
4. Traffic fluctuates based on application usage.
5. Network is bursty rather than constant.

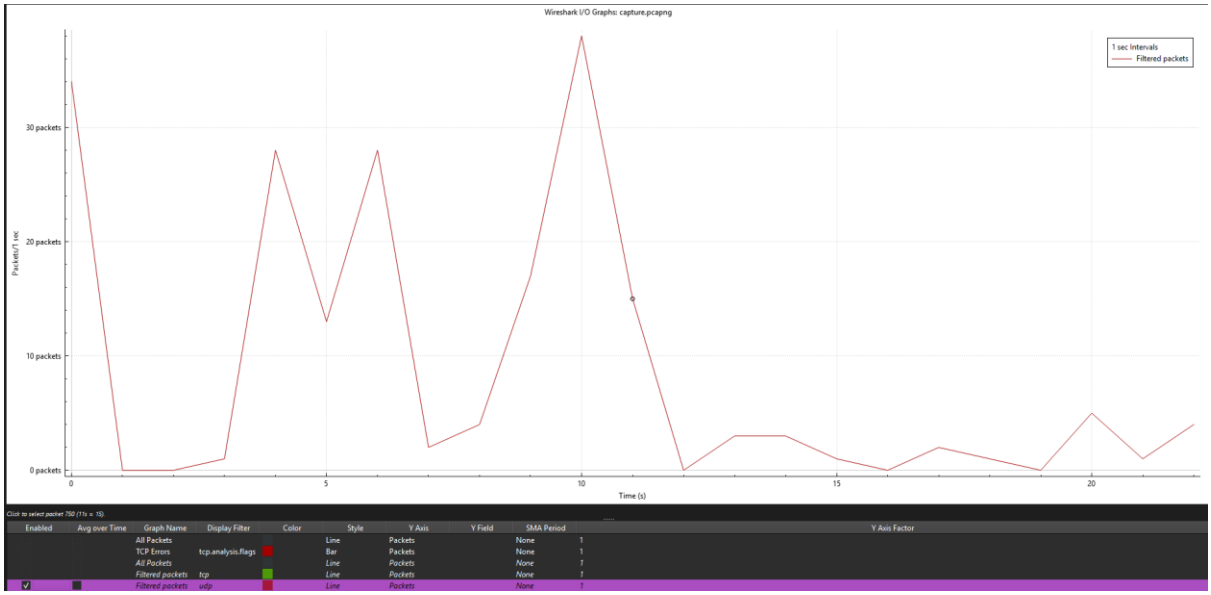
2.TCP



Inferences:

1. Majority of traffic is TCP-based.
2. Web browsing generates TCP packets.
3. TCP traffic shows congestion control behavior.
4. Spikes indicate file transfer or streaming.
5. TCP dominates compared to UDP in normal browsing.

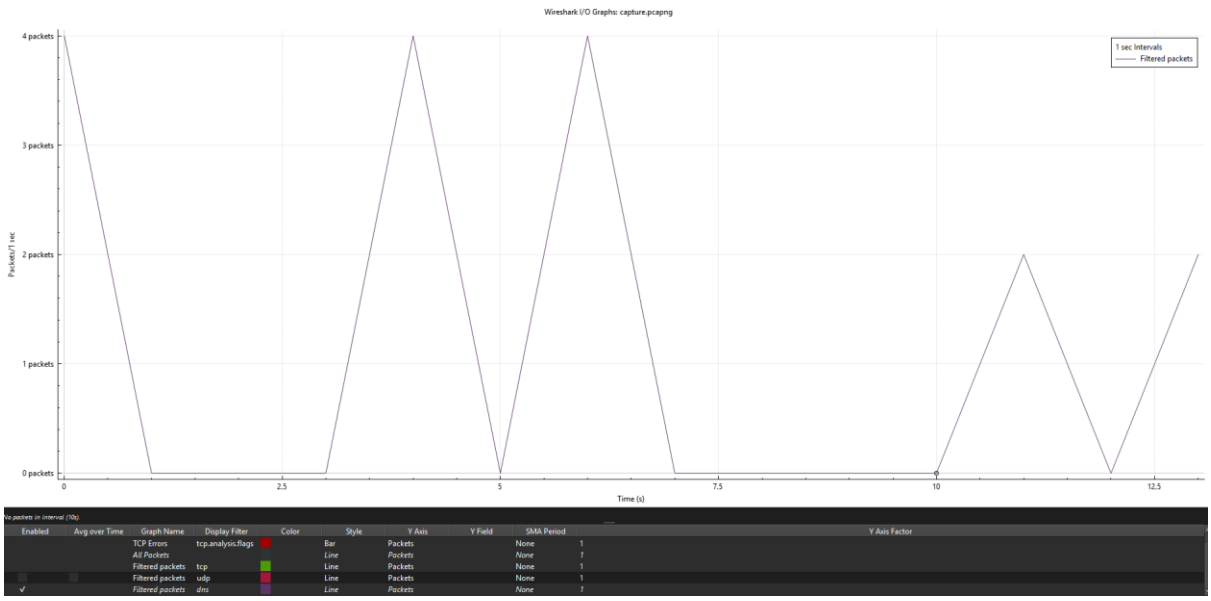
3.UDP:



Inferences:

1. UDP traffic is minimal compared to TCP.
2. DNS queries use UDP.
3. Some streaming packets use UDP.
4. Traffic is low latency but non-reliable.
5. UDP appears in short bursts.

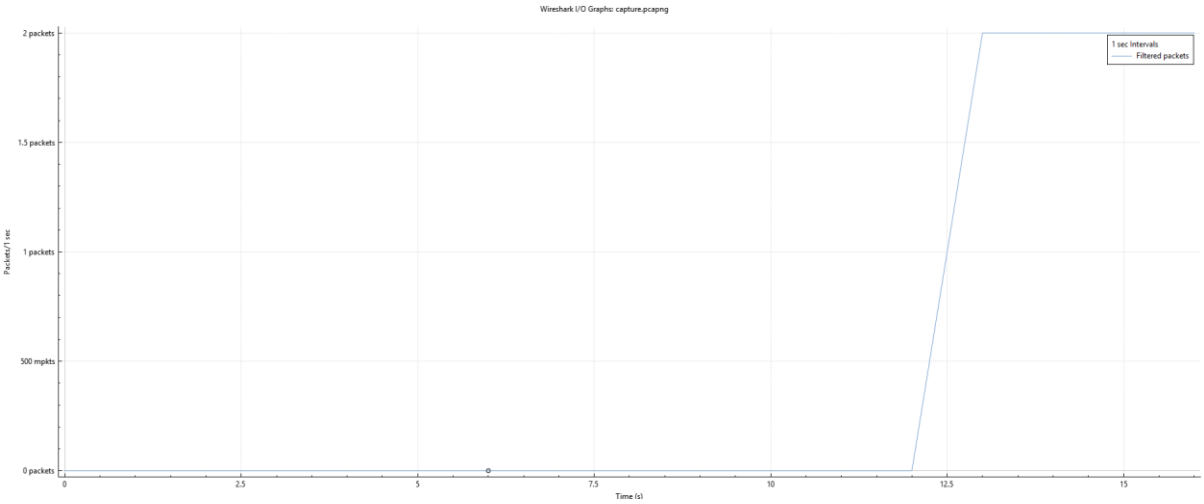
<u>4.DNS</u>:



Inferences:

1. DNS packets appear only at connection initiation.
2. Very small packet sizes.
3. DNS spikes correspond to new website visits.
4. DNS traffic is short duration.
5. Few packets compared to TCP.

5.ICMP:

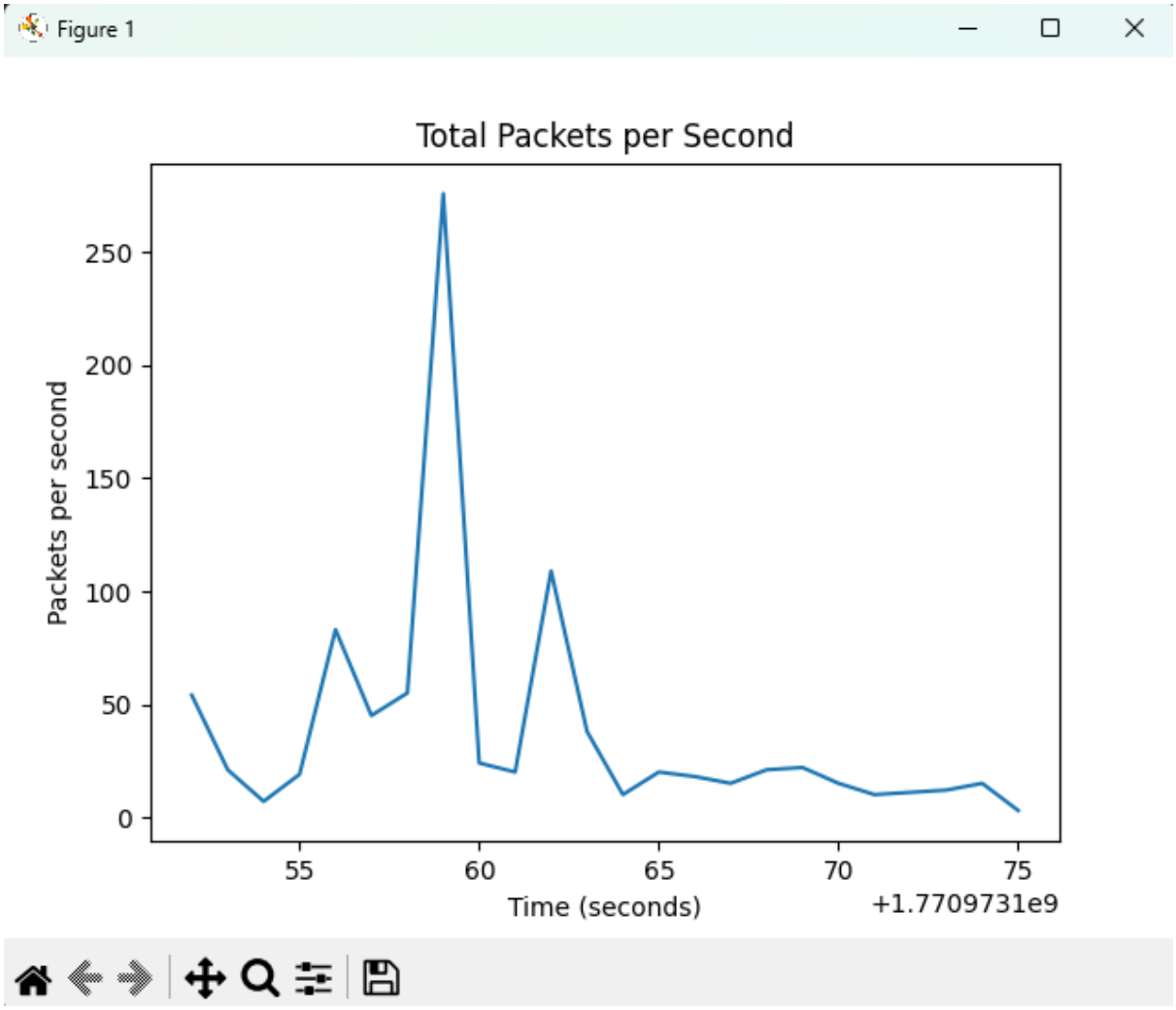


Inferences:

1. ICMP traffic appears in regular intervals during ping execution.
2. Packet rate is low compared to TCP and UDP traffic.
3. ICMP packets are mainly Echo Request and Echo Reply.
4. Traffic is symmetric between source and destination.
5. ICMP is primarily used for network diagnostics and connectivity testing.

Same Graphs using python+scrapy:

1.Total Packets:



2.tcp

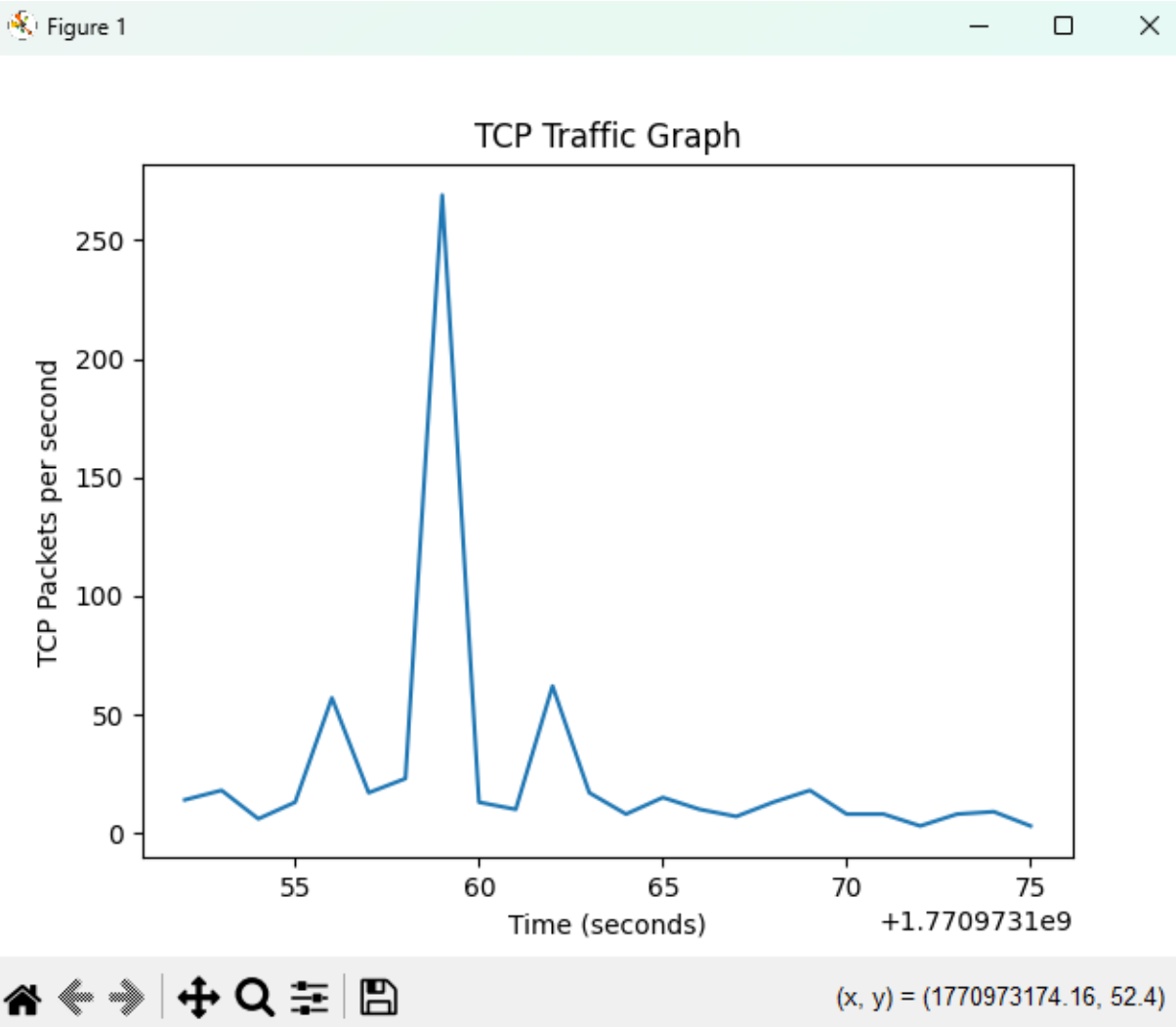


Figure 1
TCP Traffic Graph
TCP Packets per second
250
200
150
100
50
0
55
60
65
70
75
Time (seconds)
+1.7709731e9
(x, y) = (1770973174.16, 52.4)

3.udp

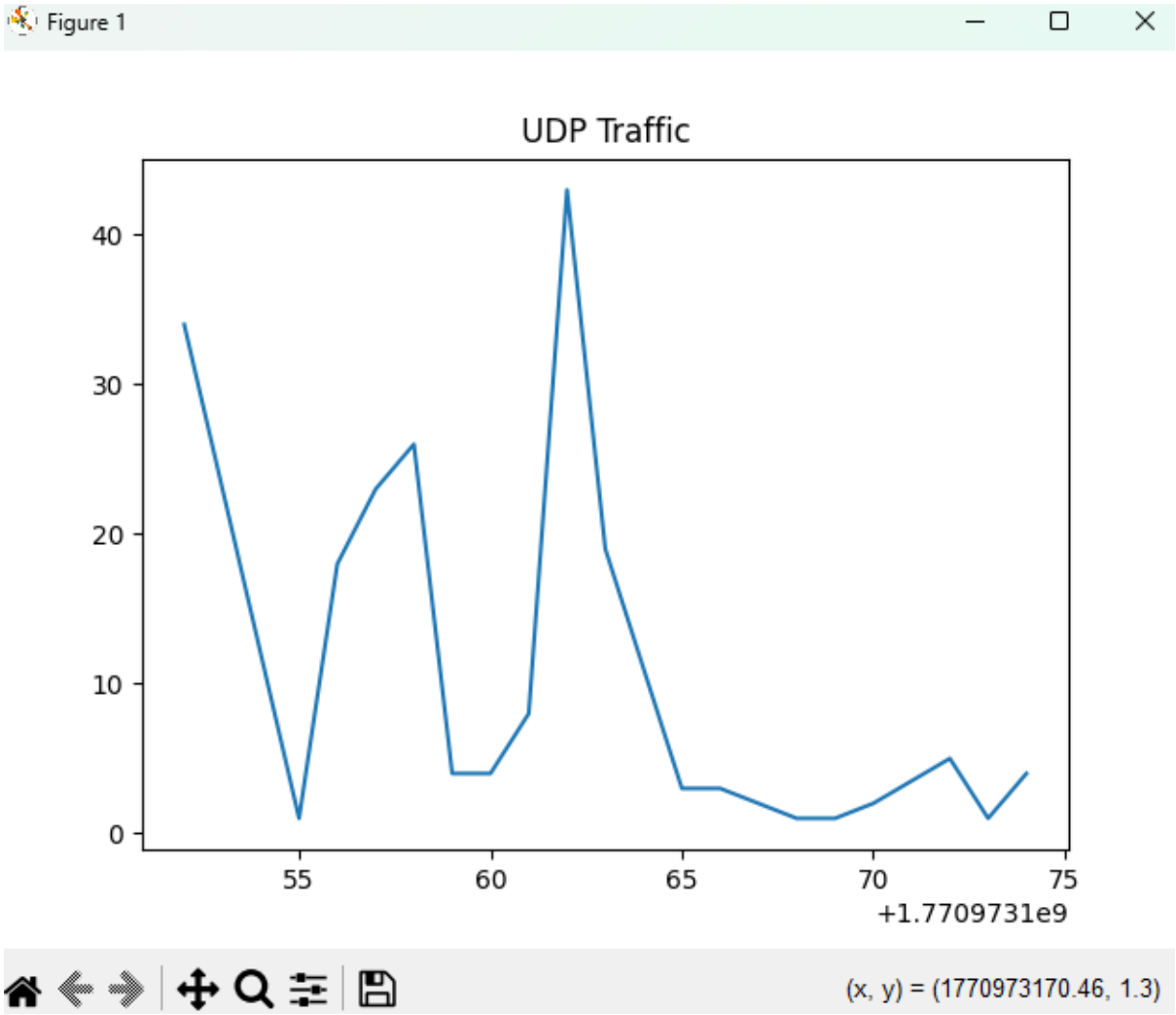


Figure 1
UDP Traffic
40
30
20
10
0
55
60
65
70
75
+1.7709731e9
(x, y) = (1770973170.46, 1.3)

4.dns:

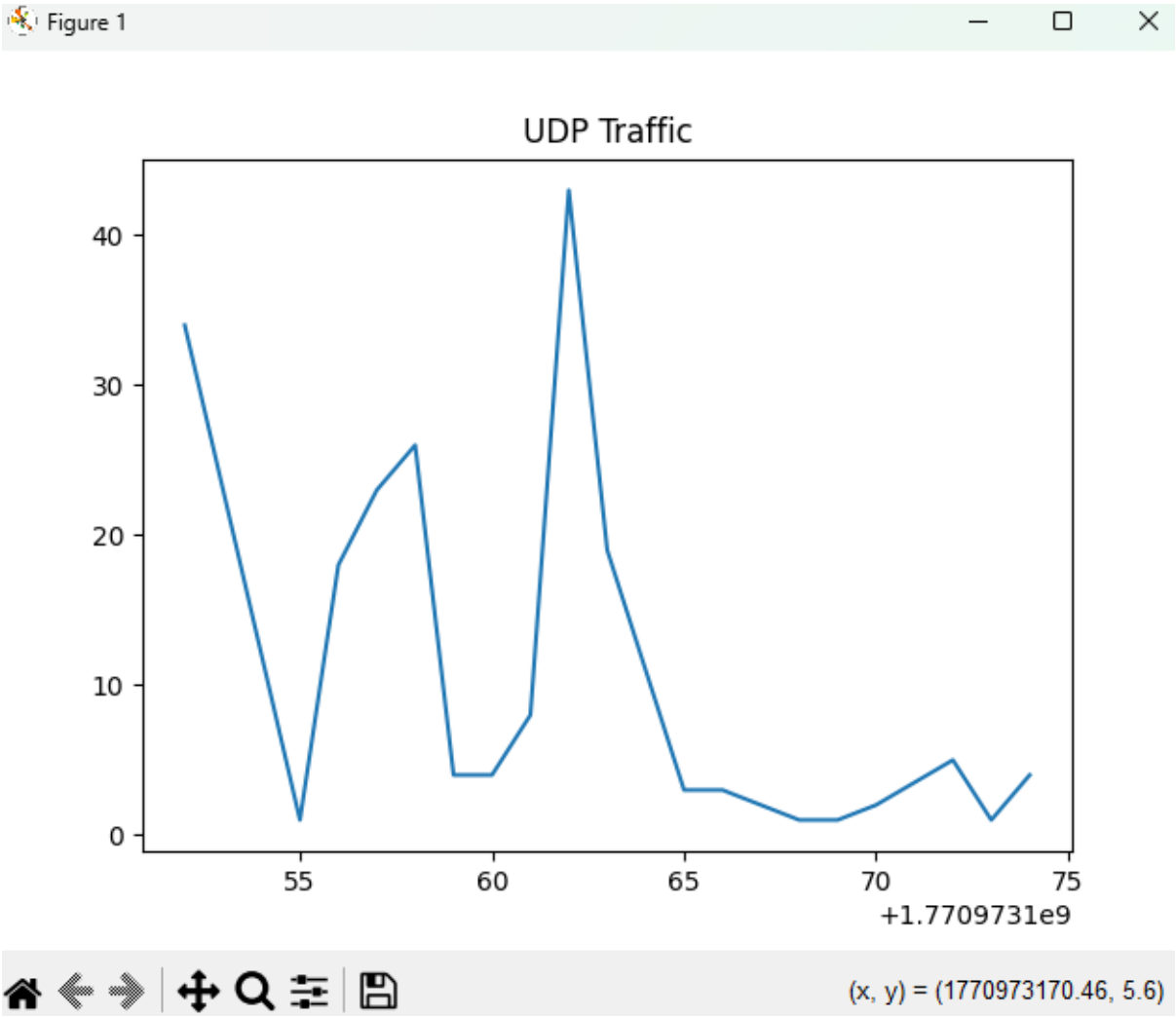


Figure 1
UDP Traffic
40
30
20
10
0
55
60
65
70
75
+1.7709731e9
(x, y) = (1770973170.46, 5.6)

5.icmp:

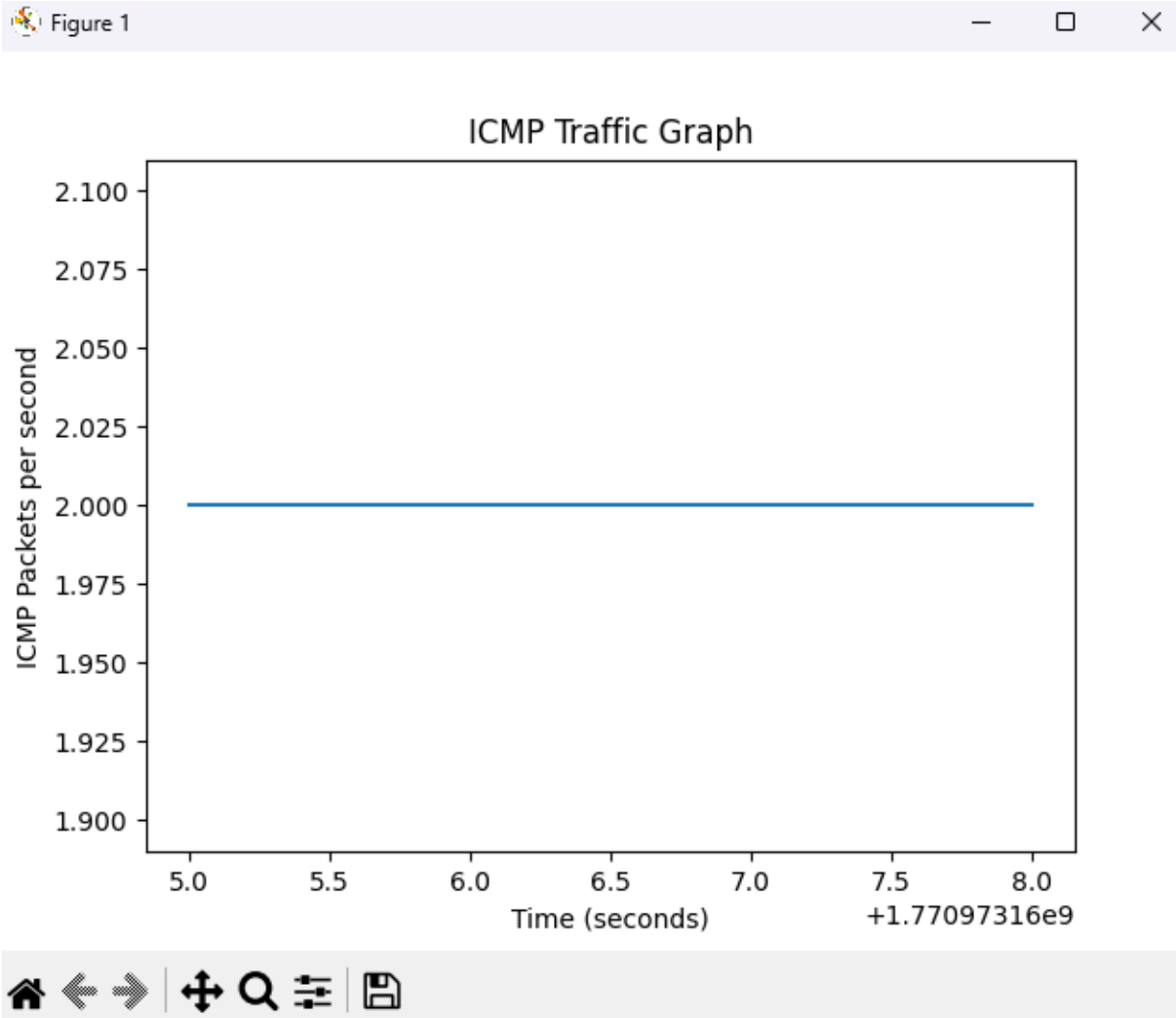


Figure 1
ICMP Traffic Graph
ICMP Packets per second
2.100
2.075
2.050
2.025
2.000
1.975
1.950
1.925
1.900
5.0
5.5
6.0
6.5
7.0
7.5
8.0
+1.77097316e9
Time (seconds)

Protocol Conversation:

OUTPUT:

```
PS C:\Users\Student\Downloads> python protocol.py
Total packets: 923
Total data size: 459819 bytes
Total header size: 16552 bytes
PS C:\Users\Student\Downloads>
```

CODE:

```
from scapy.all import *

from collections import defaultdict

packets = rdpcap("capture.pcapng")

total_packets = len(packets)

total_size = sum(len(pkt) for pkt in packets)


header_size = 0

for pkt in packets:

    if IP in pkt:

        header_size += pkt[IP].ihl * 4


print("Total packets:", total_packets)

print("Total data size:", total_size, "bytes")

print("Total header size:", header_size, "bytes")
```

Conversation printing using scrapy:

Analysis.py:

```
from scapy.all import *

from collections import defaultdict


packets = rdpcap("capture.pcap")


pair_bytes = defaultdict(int)

pair_times = defaultdict(list)

pair_protocol_packets = defaultdict(int)


for pkt in packets:

    if IP in pkt:

        src = pkt[IP].src

        dst = pkt[IP].dst

        pair = (src, dst)


        pair_bytes[pair] += len(pkt)

        pair_times[pair].append(pkt.time)


        if TCP in pkt:

            proto = "TCP"

        elif UDP in pkt:

            proto = "UDP"

        elif ICMP in pkt:
```

```
            proto = "ICMP"
        else:
            proto = "OTHER"

        pair_proto = (src, dst, proto)
        pair_protocol_packets[pair_proto] += 1

max_pair = max(pair_bytes, key=pair_bytes.get)

print("Address Pair Communicating Maximum Bytes")
print("Pair:", max_pair)
print("Total Bytes:", pair_bytes[max_pair])

print("\nAverage Inter-Packet Time Difference")

for pair in pair_times:
    times = sorted(pair_times[pair])
    if len(times) > 1:
        diffs = [t2 - t1 for t1, t2 in zip(times[:-1], times[1:])]
        avg_diff = sum(diffs) / len(diffs)
        print("Pair:", pair, "| Average Time Difference:", avg_diff, "seconds")

print("\nTotal Packets Per Pair Per Protocol")

for key, value in pair_protocol_packets.items():
```

```
    src, dst, proto = key

    print("Pair:", (src, dst), "| Protocol:", proto, "| Packets:", value)
```

OUTPUT:

```
PS C:\Users\Student\Downloads> python analysis.py

--- Address Pair Communicating Maximum Bytes ---
Pair: ('104.18.21.213', '172.16.15.123')
Total Bytes: 242107

--- Average Inter-Packet Time Difference ---
Pair: ('172.16.97.123', '172.16.15.3') | Average Time Difference: 6.0087205 seconds
Pair: ('172.16.15.123', '172.16.1.11') | Average Time Difference: 0.7816241764705882352941176471 seconds
Pair: ('172.16.9.165', '172.16.15.40') | Average Time Difference: 3.7562935 seconds
Pair: ('172.16.1.11', '172.16.15.123') | Average Time Difference: 0.885884 seconds
Pair: ('172.16.15.123', '142.250.183.174') | Average Time Difference: 0.008205647058823529411764705882 seconds
Pair: ('142.250.183.174', '172.16.15.123') | Average Time Difference: 0.014966909090909090909090909090909 seconds
Pair: ('172.16.88.64', '172.16.15.3') | Average Time Difference: 4.670253 seconds
Pair: ('172.16.73.221', '172.16.15.123') | Average Time Difference: 0.00224475 seconds
Pair: ('172.16.15.123', '172.16.73.221') | Average Time Difference: 0.004464 seconds
Pair: ('172.16.89.5', '172.16.15.3') | Average Time Difference: 6.0068015 seconds
Pair: ('172.16.96.109', '172.16.15.83') | Average Time Difference: 4.676523333333333333333333333 seconds
Pair: ('172.16.8.79', '172.16.15.9') | Average Time Difference: 8.010071 seconds
Pair: ('172.16.9.207', '172.16.15.123') | Average Time Difference: 0.0011195 seconds
Pair: ('172.16.15.123', '172.16.9.207') | Average Time Difference: 0.0021995 seconds
Pair: ('172.16.10.134', '172.16.15.11') | Average Time Difference: 5.004554666666666666666666667 seconds
Pair: ('172.16.38.182', '172.16.15.40') | Average Time Difference: 8.001011 seconds
Pair: ('172.16.75.156', '172.16.15.9') | Average Time Difference: 6.004221 seconds
Pair: ('172.16.25.233', '172.16.15.83') | Average Time Difference: 6.0089005 seconds
Pair: ('172.16.80.169', '172.16.15.9') | Average Time Difference: 6.002050 seconds
Pair: ('172.16.10.134', '172.16.15.3') | Average Time Difference: 6.006850 seconds
Pair: ('172.16.88.107', '172.16.15.3') | Average Time Difference: 6.0079845 seconds
Pair: ('172.16.96.145', '172.16.15.3') | Average Time Difference: 3.7593535 seconds
Pair: ('172.16.88.127', '172.16.15.40') | Average Time Difference: 3.7606065 seconds
Pair: ('172.16.80.109', '172.16.15.145') | Average Time Difference: 3.75388275 seconds
Pair: ('172.16.25.96', '172.16.15.83') | Average Time Difference: 3.7647205 seconds
Pair: ('172.16.97.161', '172.16.15.11') | Average Time Difference: 8.004955 seconds
Pair: ('172.16.9.48', '172.16.15.145') | Average Time Difference: 8.002668 seconds
Pair: ('0.0.0.0', '255.255.255.255') | Average Time Difference: 2.1156864444444444444444444444 seconds
Pair: ('172.16.25.94', '172.16.15.9') | Average Time Difference: 8.002224 seconds
Pair: ('172.16.9.41', '172.16.15.3') | Average Time Difference: 8.005992 seconds
Pair: ('172.16.25.229', '172.16.15.3') | Average Time Difference: 8.004914 seconds
Pair: ('172.16.80.35', '172.16.15.3') | Average Time Difference: 3.7656475 seconds
Pair: ('172.16.11.39', '172.16.15.100') | Average Time Difference: 3.75746825 seconds
Pair: ('172.16.15.123', '40.126.17.132') | Average Time Difference: 0.036859285714285714285714285571 seconds
Pair: ('40.126.17.132', '172.16.15.123') | Average Time Difference: 0.019817307692307692307692307769 seconds
Pair: ('172.16.88.41', '172.16.15.123') | Average Time Difference: 6.114477 seconds
Pair: ('172.16.15.123', '172.16.88.41') | Average Time Difference: 6.017912 seconds
Pair: ('172.16.15.123', '23.65.124.99') | Average Time Difference: 0.190677542857142857142857142
```

```
Pair: ('172.16.56.38', '172.16.15.123') | Average Time Difference: 5.065924 seconds
Pair: ('172.16.15.123', '172.16.56.38') | Average Time Difference: 4.969128 seconds
Pair: ('172.16.25.6', '172.16.15.123') | Average Time Difference: 4.884564 seconds
Pair: ('172.16.15.123', '172.16.25.6') | Average Time Difference: 4.774954 seconds
Pair: ('172.16.97.231', '172.16.15.3') | Average Time Difference: 3.75892975 seconds
Pair: ('172.16.101.59', '224.0.0.252') | Average Time Difference: 0.016465 seconds
Pair: ('172.16.15.146', '224.0.0.252') | Average Time Difference: 0.411729 seconds
Pair: ('172.16.9.30', '172.16.15.123') | Average Time Difference: 4.356392 seconds
Pair: ('172.16.15.123', '172.16.9.30') | Average Time Difference: 4.263551 seconds
Pair: ('172.16.25.21', '172.16.15.123') | Average Time Difference: 4.309794 seconds
Pair: ('172.16.75.131', '172.16.15.9') | Average Time Difference: 3.760240 seconds
Pair: ('172.16.15.123', '172.16.25.21') | Average Time Difference: 4.216312 seconds
Pair: ('172.16.9.160', '172.16.15.123') | Average Time Difference: 4.240533 seconds
Pair: ('172.16.15.123', '172.16.9.160') | Average Time Difference: 4.137283 seconds
Pair: ('172.16.15.123', '185.125.190.21') | Average Time Difference: 0.03785634615384615384615384615 seconds
Pair: ('185.125.190.21', '172.16.15.123') | Average Time Difference: 0.020500041666666666666666666667 seconds
Pair: ('172.16.89.7', '172.16.15.3') | Average Time Difference: 3.758610 seconds
Pair: ('172.16.15.123', '104.18.21.213') | Average Time Difference: 0.00356348 seconds
Pair: ('104.18.21.213', '172.16.15.123') | Average Time Difference: 0.000551714285714285714285
```

```
--- Total Packets Per Pair Per Protocol ---
Pair: ('172.16.97.123', '172.16.15.3') | Protocol: TCP | Packets: 3
Pair: ('172.16.15.123', '172.16.1.11') | Protocol: UDP | Packets: 6
Pair: ('172.16.9.165', '172.16.15.40') | Protocol: TCP | Packets: 5
Pair: ('172.16.1.11', '172.16.15.123') | Protocol: UDP | Packets: 6
Pair: ('172.16.15.123', '142.250.183.174') | Protocol: UDP | Packets: 18
Pair: ('142.250.183.174', '172.16.15.123') | Protocol: UDP | Packets: 12
Pair: ('172.16.88.64', '172.16.15.3') | Protocol: TCP | Packets: 4
Pair: ('172.16.75.202', '172.16.15.3') | Protocol: TCP | Packets: 1
Pair: ('172.16.73.221', '172.16.15.123') | Protocol: TCP | Packets: 5
Pair: ('172.16.15.123', '172.16.73.221') | Protocol: TCP | Packets: 3
Pair: ('172.16.89.5', '172.16.15.3') | Protocol: TCP | Packets: 3
Pair: ('172.16.96.109', '172.16.15.83') | Protocol: TCP | Packets: 4
Pair: ('172.16.8.79', '172.16.15.9') | Protocol: TCP | Packets: 2
Pair: ('172.16.9.207', '172.16.15.123') | Protocol: TCP | Packets: 5
Pair: ('172.16.15.123', '172.16.9.207') | Protocol: TCP | Packets: 3
Pair: ('172.16.10.134', '172.16.15.11') | Protocol: TCP | Packets: 4
Pair: ('172.16.38.182', '172.16.15.40') | Protocol: TCP | Packets: 2
Pair: ('172.16.75.156', '172.16.15.9') | Protocol: TCP | Packets: 3
Pair: ('172.16.25.233', '172.16.15.83') | Protocol: TCP | Packets: 3
Pair: ('172.16.80.169', '172.16.15.9') | Protocol: TCP | Packets: 3
Pair: ('172.16.10.134', '172.16.15.3') | Protocol: TCP | Packets: 3
Pair: ('172.16.88.107', '172.16.15.3') | Protocol: TCP | Packets: 3
Pair: ('172.16.96.145', '172.16.15.3') | Protocol: TCP | Packets: 5
Pair: ('172.16.88.127', '172.16.15.40') | Protocol: TCP | Packets: 5
Pair: ('172.16.80.109', '172.16.15.145') | Protocol: TCP | Packets: 5
Pair: ('172.16.25.96', '172.16.15.83') | Protocol: TCP | Packets: 5
Pair: ('172.16.15.123', '142.250.4.188') | Protocol: TCP | Packets: 1
Pair: ('142.250.4.188', '172.16.15.123') | Protocol: TCP | Packets: 1
Pair: ('172.16.97.161', '172.16.15.11') | Protocol: TCP | Packets: 2
Pair: ('172.16.9.48', '172.16.15.145') | Protocol: TCP | Packets: 2
Pair: ('0.0.0.0', '255.255.255.255') | Protocol: UDP | Packets: 10
Pair: ('172.16.25.94', '172.16.15.9') | Protocol: TCP | Packets: 2
Pair: ('172.16.75.187', '172.16.15.9') | Protocol: TCP | Packets: 1
Pair: ('172.16.9.41', '172.16.15.3') | Protocol: TCP | Packets: 2
Pair: ('172.16.80.124', '172.16.15.3') | Protocol: TCP | Packets: 1
Pair: ('172.16.25.229', '172.16.15.3') | Protocol: TCP | Packets: 2
Pair: ('172.16.80.35', '172.16.15.3') | Protocol: TCP | Packets: 5
Pair: ('172.16.96.101', '172.16.15.145') | Protocol: TCP | Packets: 1
Pair: ('172.16.11.39', '172.16.15.100') | Protocol: TCP | Packets: 5
Pair: ('172.16.15.123', '40.126.17.132') | Protocol: TCP | Packets: 8
Pair: ('40.126.17.132', '172.16.15.123') | Protocol: TCP | Packets: 14
Pair: ('172.16.88.41', '172.16.15.123') | Protocol: TCP | Packets: 2
Pair: ('172.16.15.123', '172.16.88.41') | Protocol: TCP | Packets: 2
Pair: ('172.16.96.141', '172.16.15.83') | Protocol: TCP | Packets: 1
Pair: ('172.16.15.123', '172.16.1.11') | Protocol: TCP | Packets: 12
Pair: ('172.16.1.11', '172.16.15.123') | Protocol: TCP | Packets: 10
Pair: ('172.16.15.123', '23.65.124.99') | Protocol: UDP | Packets: 36
Pair: ('172.16.9.164', '172.16.15.3') | Protocol: TCP | Packets: 1
Pair: ('172.16.97.198', '172.16.15.3') | Protocol: TCP | Packets: 4
```

```
Pair: ('172.16.88.23', '172.16.15.123') | Protocol: TCP | Packets: 2
Pair: ('172.16.88.33', '172.16.15.123') | Protocol: TCP | Packets: 1
Pair: ('172.16.76.26', '172.16.15.123') | Protocol: TCP | Packets: 1
Pair: ('172.16.89.167', '172.16.15.123') | Protocol: TCP | Packets: 1
Pair: ('172.16.80.149', '172.16.15.123') | Protocol: TCP | Packets: 1
Pair: ('172.16.38.143', '172.16.15.123') | Protocol: TCP | Packets: 1
Pair: ('172.16.89.62', '172.16.15.123') | Protocol: TCP | Packets: 1
Pair: ('172.16.9.44', '172.16.15.123') | Protocol: TCP | Packets: 1
Pair: ('172.16.80.153', '172.16.15.123') | Protocol: TCP | Packets: 1
Pair: ('172.16.96.48', '172.16.15.123') | Protocol: TCP | Packets: 2
Pair: ('172.16.89.27', '172.16.15.123') | Protocol: TCP | Packets: 1
Pair: ('172.16.89.75', '172.16.15.123') | Protocol: TCP | Packets: 1
Pair: ('172.16.96.158', '172.16.15.123') | Protocol: TCP | Packets: 1
Pair: ('172.16.88.75', '172.16.15.123') | Protocol: TCP | Packets: 1
Pair: ('172.16.76.29', '172.16.15.123') | Protocol: TCP | Packets: 2
Pair: ('172.16.97.243', '172.16.15.3') | Protocol: TCP | Packets: 3
Pair: ('172.16.88.105', '172.16.15.3') | Protocol: TCP | Packets: 4
Pair: ('172.16.75.201', '172.16.15.3') | Protocol: TCP | Packets: 4
Pair: ('172.16.24.249', '172.16.15.40') | Protocol: TCP | Packets: 2
Pair: ('172.16.9.169', '172.16.15.40') | Protocol: TCP | Packets: 3
Pair: ('172.16.15.198', '172.16.15.255') | Protocol: UDP | Packets: 1
Pair: ('172.16.88.148', '172.16.15.3') | Protocol: TCP | Packets: 4
Pair: ('172.16.15.123', '142.251.222.206') | Protocol: ICMP | Packets: 4
Pair: ('142.251.222.206', '172.16.15.123') | Protocol: ICMP | Packets: 4
Pair: ('172.16.56.24', '172.16.15.9') | Protocol: TCP | Packets: 2
Pair: ('172.16.96.121', '172.16.15.145') | Protocol: TCP | Packets: 4
Pair: ('172.16.15.111', '172.16.15.255') | Protocol: UDP | Packets: 1
Pair: ('172.16.26.207', '172.16.15.3') | Protocol: TCP | Packets: 3
Pair: ('172.16.96.25', '172.16.15.3') | Protocol: TCP | Packets: 3
Pair: ('172.16.80.177', '172.16.15.37') | Protocol: TCP | Packets: 4
Pair: ('172.16.97.126', '172.16.15.3') | Protocol: TCP | Packets: 2
Pair: ('172.16.89.140', '172.16.15.3') | Protocol: TCP | Packets: 3
Pair: ('172.16.89.129', '172.16.15.3') | Protocol: TCP | Packets: 3
Pair: ('172.16.26.164', '172.16.15.83') | Protocol: TCP | Packets: 2
Pair: ('172.16.15.123', '172.16.10.224') | Protocol: TCP | Packets: 5
Pair: ('172.16.10.224', '172.16.15.123') | Protocol: TCP | Packets: 3
Pair: ('172.16.15.116', '172.16.15.255') | Protocol: UDP | Packets: 1
Pair: ('172.16.25.234', '172.16.15.3') | Protocol: TCP | Packets: 2
Pair: ('172.16.25.160', '172.16.15.3') | Protocol: TCP | Packets: 3
Pair: ('172.16.15.96', '172.16.15.255') | Protocol: UDP | Packets: 1
Pair: ('172.16.15.132', '172.16.15.255') | Protocol: UDP | Packets: 1
Pair: ('172.16.15.95', '172.16.15.255') | Protocol: UDP | Packets: 1
Pair: ('172.16.75.132', '172.16.15.9') | Protocol: TCP | Packets: 2
Pair: ('172.16.15.123', '172.16.15.255') | Protocol: UDP | Packets: 1
Pair: ('172.16.96.154', '172.16.15.9') | Protocol: TCP | Packets: 1
Pair: ('172.16.15.142', '172.16.15.255') | Protocol: UDP | Packets: 1
Pair: ('172.16.15.54', '224.0.0.251') | Protocol: UDP | Packets: 1
Pair: ('172.16.25.98', '172.16.15.3') | Protocol: TCP | Packets: 1
Pair: ('172.16.26.225', '172.16.15.37') | Protocol: TCP | Packets: 1
Pair: ('172.16.97.150', '172.16.15.3') | Protocol: TCP | Packets: 1
```